

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 When two ideal polarizers are placed with their transmission axes at 90° the transmitted intensity is.....
 a. Maximum b. Half of the incident intensity
 c. Zero d. One-fourth of the incident intensity
- 2 An optical fiber has a numerical aperture of 0.5. The acceptance angle in air is
 a. 15° b. 30° c. 45° d. 50°
- 3 For the operator d/dx the function e^{2x} has eigenvalue.....
 a. 1 b. 2 c. 6 d. e^{2x}
- 4 A qubit can exist in....
 a. State 0 only b. State 1 only
 c. State 2 only d. Superposition of 0 and 1
- 5 In an electromagnetic wave, the electric field and magnetic field are
 a. Parallel b. Perpendicular
 c. Opposite d. Unrelated
- 6 Polar molecules possess.....
 a. Only induced dipole moment b. No dipole moment
 c. Permanent dipole moment d. Zero charge
- 7 Among α , β and γ radiations, the highest ionising power is possessed by....
 a. γ -rays b. α -rays c. β -rays d. all are equal
- 8 Miller indices are used to represent
 a. Crystal defects b. Crystal planes
 c. Crystal density d. Crystal systems
- 9 The coordination number of FCC structure is
 a. 12 b. 8 c. 4 d. 6
- 10 Which synthesis method belongs to the top-down approach
 a. Sol-gel method b. Chemical vapor deposition
 c. Ball milling d. Self-assembly
- 11 The Hall Effect occurs due to the action of
 a. Electric force only b. Magnetic force on moving charge carriers
 c. Gravitational force d. Nuclear force.

- 12 VSM stands for
 a. Variable Sample Magnetometer b. Vector Sample Meter
 c. Voltage Sensing Magnetometer d. Vibrating Sample Magnetometer

Q2. Solve the following.

- A) Explain stimulated emission with diagram. In a Newton's rings experiment, the radius of curvature of the lens 6

is 2 m. The diameter of the 15th dark ring is 5.0 mm. Calculate the wavelength of light used.

- B) What is forced oscillation? Obtain differential equation for forced oscillation. 6

Q3. Solve the following.

- A) Derive the expression for the energy eigenvalues of a particle confined in a one-dimensional infinite potential box of length L. 6

- B) Derive Schrodinger time dependent wave equation. 6

Q4. Solve Any Two of the following.

- A) What is dielectric polarization explain with diagram. Explain orientation polarization. 6

- B) What is Del operator? Determine Curl ($\nabla \times F$) of following vector function 6

$$\vec{F}(x, y, z) = (xy + z)\hat{i} + (x^2 - yz)\hat{j} + (xz + y)\hat{k}$$

- C) Derive the electromagnetic wave equation in free space from Maxwell's equations and show that electromagnetic waves travel with the speed of light. 6

Q5. Solve Any Two of the following. <https://www.batuonline.com>

- A) Define unit cell. Find atomic radius for BCC and FCC with suitable diagram. 6

- B) What is radioactivity and write the properties of alpha beta and gamma rays. 6

- C) Draw crystal plane having miller indices (110) and (111). Calculate the density of copper having FCC structure if the atomic weight is 63.5 g/mol, lattice parameter is 3.61×10^{-8} cm, and Avogadro's number is 6.023×10^{23} atoms/mol. 6

Q6. Solve Any Two of the following.

- A) What is superconductivity? Differentiate between type I and type II superconductor. 6

- B) Explain diamagnetic, paramagnetic, ferromagnetic materials with suitable diagram. 6

- C) Derive an expression for conductivity of semiconductor. 6

A semiconductor has carrier concentration of $5 \times 10^{22} \text{ m}^{-3}$. Calculate the Hall coefficient.

*** End ***